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DATA SCIENCE MINOR PROJECT REPORT

U.S. Border Crossing Entry Analysis (1996 - 2025)

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1. INTRODUCTION

The movement of people and goods across international borders is a primary indicator of economic health, tourism trends, and geopolitical stability. This project focuses on the **U.S. Border Crossing Entry Data**, which provides a detailed account of vehicles, equipment, and persons entering the United States from Canada and Mexico.

The objective of this analysis is to identify long-term trends, seasonal patterns, and the impact of regional logistics on border traffic. By utilizing Data Science techniques and Power BI visualization, this report transforms over 27 years of raw data into actionable insights regarding port utilization and transport modes.

2. SOURCE OF DATASET

The dataset was sourced from **Data.gov (U.S. General Services Administration)**, provided by the **Bureau of Transportation Statistics (BTS)**.

- **URL:** <https://catalog.data.gov/dataset/border-crossing-entry-data>
- **Dataset Size:** 273,391 Records.
- **Attributes:** Port Name, State, Port Code, Border, Date, Measure, Value, Latitude, Longitude, and Point.

3. DATASET PREPROCESSING

Before analysis, the raw CSV data underwent several preprocessing steps to ensure accuracy:

1. **Date Conversion:** The Date column (initially text like "Jan 2024") was converted into a standard DateTime format using Power Query.
2. **Handling Missing Values:** Minor null values in geospatial columns (Latitude/Longitude) were cross-referenced with Port Codes to maintain geographic integrity.
3. **Categorical Grouping:** To simplify analysis, the eight original Measure types were grouped into three logical categories using a DAX SWITCH function:
 - **Commercial:** Trucks, Trains.
 - **Private:** Personal Vehicles, Personal Vehicle Passengers, Pedestrians.
 - **Public:** Buses, Bus Passengers, Train Passengers.
4. **Creating a Calendar Table:** A custom DAX Calendar table was built to enable Year-over-Year (YoY) and Month-over-Month (MoM) comparisons.

4. ANALYSIS ON DATASET

General Description

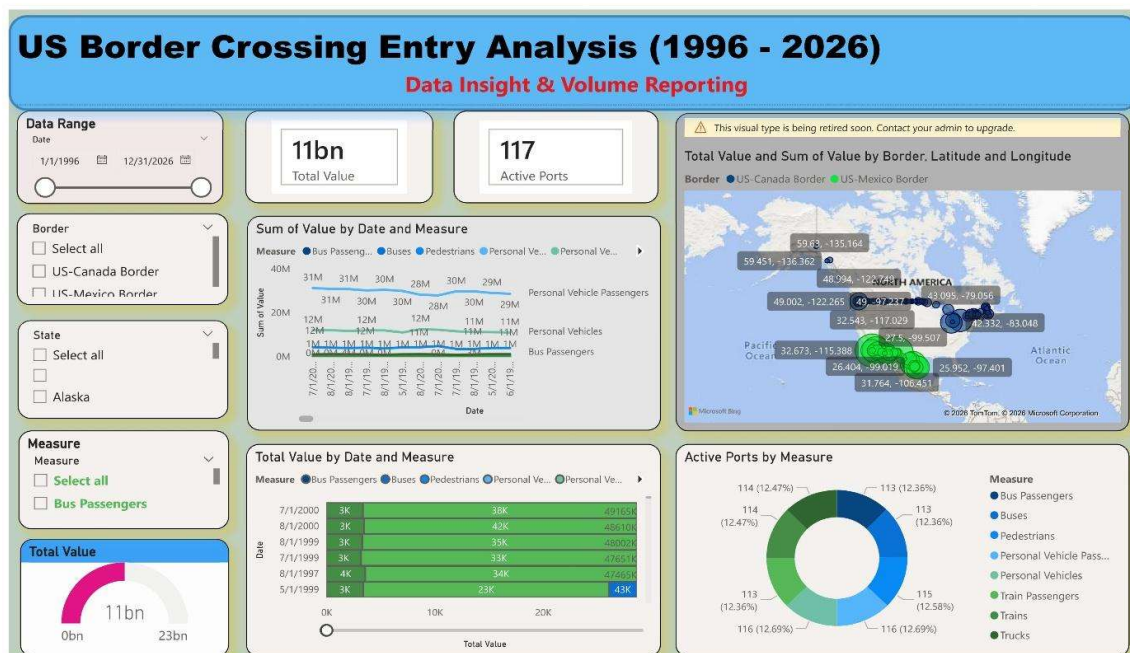
The data spans from **April 1996 to September 2025**, providing a comprehensive view of North American cross-border activity. The dataset is categorical in nature (States, Borders, Measures) with a primary numerical metric: Value (the count of entries).

Specific Requirements

- Identify the peak traffic periods.
- Compare the volume between the US-Mexico and US-Canada borders.
- Analyze the busiest states and their primary modes of transport.
- Identify seasonal fluctuations in pedestrian vs. vehicle traffic.

Analysis Results

- **The Peak Year:** Through DAX analysis, the year **2000** was identified as the all-time peak for border entries, recording over **527 Million** total crossings.
- **Border Dominance:** The **US-Mexico Border** consistently accounts for roughly **65%** of total entry volume, driven heavily by personal vehicles and pedestrian commuters.
- **State Analysis:** **Texas** emerged as the highest-volume state, acting as the primary gateway for international trade and travel.

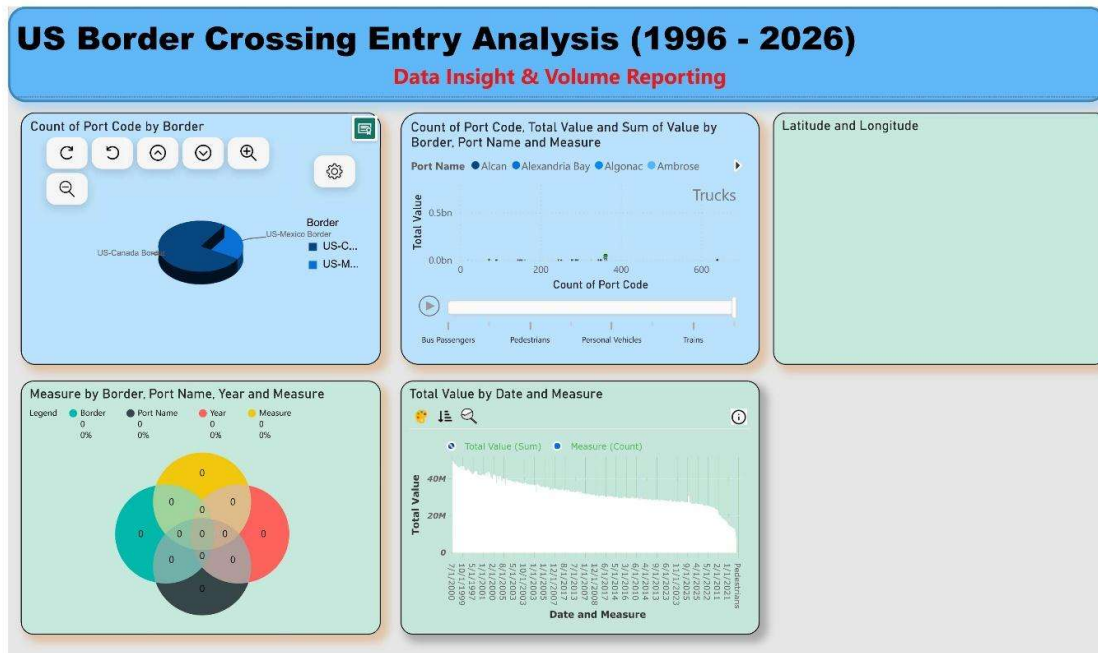


Visualization

The project utilized a Power BI Dashboard featuring:

1. **Map Visual:** Bubble map showing port density using Latitude/Longitude.
2. **Trend Lines:** Multi-line charts showing the divergence of Commercial vs. Private traffic over time.
3. **KPI Cards:** Dynamic cards displaying Total Value, Active Ports (117), and the Peak Year.

4. **Slicers:** Interactive filters for Border, State, and Grouping (Commercial/Private/Public).



5. CONCLUSION

The analysis concludes that while total border entries peaked at the turn of the millennium (2000), the nature of crossings has shifted. Commercial traffic (Trucks/Trains) has remained resilient and integrated, whereas passenger traffic is more susceptible to policy changes and global events. The dashboard effectively serves as a monitoring tool for infrastructure and logistics planning.

6. FUTURE SCOPE

- **Economic Correlation:** Integrating GDP or trade value data to see how crossing volumes correlate with economic growth.
- **Predictive Modeling:** Using Python (Scikit-learn) to forecast future traffic volumes for the next 5 years based on the current 27-year trend.
- **External Factors:** Overlaying the data with weather patterns or fuel prices to analyze their impact on pedestrian and personal vehicle choices.

7. REFERENCES

1. Bureau of Transportation Statistics. (2024). *Border Crossing Entry Data*. Data.gov.
 2. Microsoft Power BI Documentation. *DAX and Time Intelligence Functions*.
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